Ricky Faure

Principal Software Engineer

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SUMMARY

I build enterprise-grade platform infrastructure with Kubernetes Operators, Go, and Terraform. Principal Software Engineer at Optum, leading Kafka-as-a-Service and Elasticsearch-as-a-Service platforms that move 20+ petabytes of data across 800+ clusters at five nines reliability.

SKILLS

Core: Go | Kubernetes | Kubernetes Operators | Kubebuilder / Operator SDK | Terraform | GCP | GKE | Apache Kafka | Elasticsearch | Helm | CI/CD | Python | distributed systems | SRE | infrastructure as code

Additional: Docker | Prometheus | Grafana | GitOps | GitHub Actions | bare-metal Kubernetes | certificate management | Java | Spring Boot | AI-assisted development | Claude Code | mentorship and technical teaching

EXPERIENCE

Optum, UnitedHealth Group

JAN 2020 — PRESENT

Principal Software Engineer, TLCP — Technology Development Program Associate (Jan 2020) → Software Engineer (Jan 2021) → Senior Software Engineer (Mar 2022) → Lead Software Engineer (Feb 2023) → Principal Software Engineer, TLCP (Feb 2024 - Present)

Empower hundreds of teams across the enterprise to build data-driven services, moving tens of petabytes of data with minimal architectural overhead, using Kubernetes Operators and CI/CD to automate deployment, management, and hosting of enterprise-grade distributed systems across on-premise and multi-cloud environments.

- Platform and team leadership: Lead 15 engineers across the Kafka-as-a-Service and Elasticsearch-as-a-Service platforms and advise 2 engineering leaders , sustaining five nines reliability through Kubernetes Operators and disciplined infrastructure-as-code practices.
- Kubernetes Operator development: Build and maintain custom Kubernetes Operators (Go, Kubebuilder) for Kafka, Elasticsearch, Prometheus, Service Monitors, GCP VolumeSnapshots, certificate management, and bare-metal Kubernetes upgrade orchestration.
- Control plane architecture: Design and manage the platform's Kubernetes resource management control plane, managing 800+ clusters across thousands of nodes in multi-tenant on-premise and GCP environments with zero customer data loss in production.
- Self-service marketplace migration: Achieved a 52% increase in resource deployments by migrating from GitOps to a GUI-based management system within an enterprise marketplace, backed by a custom Terraform Provider and Kubernetes resource manager.
- Cloud cost optimization: Delivered $2.5M in annual cost savings through cloud resource optimization, instance type migrations, and elimination of excessive log retention in GCP.
- Cloud scale and GKE orchestration: Expanded the platform to handle 20+ petabytes of data movement across on-premise and GCP by extending Kubernetes Operators for cloud environments; own provisioning and orchestration of the 30+ GKE clusters backing these platforms through Terraform.
- Agentic development workspaces: Design and build agent-driven development workspaces that standardize the team's engineering workflow into a consistent, deterministic process, with reusable agent skills that automate remediation and SRE support tasks.
- Featured Project: Warpstream Cluster Provisioning Platform (Q4 2025 - Q1 2026)
- Architected the end-to-end design for Warpstream-based Kafka cluster provisioning, from the customer-facing resource manager down to the backend infrastructure, and delivered a net-new Warpstream operator (Go) plus the Terraform cloud infrastructure behind it.

- Shipped to two of Optum's largest Apache Kafka on GCP customers as beta; Warpstream's diskless architecture is projected to reduce their annual Kafka infrastructure spend by approximately 80% .

Technologies: Go, Kubebuilder, Kubernetes Operators, Helm, Terraform, GCP, GKE, GitHub Actions, Python, Kafka, Warpstream, Elasticsearch, Prometheus, Docker

Optum

JUN — AUG, 2017 — 2019

TDP Software Development Intern

- Returned for three consecutive summer internships, contributing business-critical features to a monolithic application with automated unit and integration testing.

Technologies: Java, Spring Boot, Groovy Spock

PROJECTS

Yo-Yo Mount Visualizer (active development)

TypeScript · React · react-three-fiber · graph modeling · pathfinding · physics simulation

3D trick engine that models yo-yo string mounts as graph topologies and aims to discover new tricks through pathfinding. Encodes mounts as schema-validated string traversals with canonical hashing, rendered in an interactive 3D visualizer with a Verlet rope physics simulation.

Spin Ledger (closed source)

Python · FastAPI · SQLAlchemy 2.0 · Alembic · rule-based entity extraction · data normalization

Single pane of glass for buy/sell/trade activity across skill toy community forums. Ingests unstructured, schemaless forum listings, normalizes them into a taxonomy-aware data model via deterministic rule-based entity extraction, and serves them through an indexed, searchable marketplace dashboard.

Self-Hosted Home Lab

Docker · Caddy · reverse proxy · TLS · DNS · VPN tunneling · NAS · Linux

Multi-machine home lab orchestrating containerized services with Docker, fronted by Caddy as a reverse proxy with automatic HTTPS; every service is served exclusively over TLS. Hosts a media server, network-attached storage, network-wide DNS-sinkhole ad blocking, secrets vaults, and closed-source project deployments, with private encrypted tunnels for secure remote access.

OSS Feed

JavaScript · GitHub Actions · RSS/Atom · GitHub Pages

Self-hosted weekly RSS digest that tracks releases from curated open source projects. A watchlist YAML feeds GitHub Actions to fetch releases, generate an XML feed, and publish via GitHub Pages, with Slack/Discord webhook notifications.

Sheet Vision (Senior Design)

ElectronJS · ReactJS · AWS · Python · computer vision

Application that reads sheet music, plays it back, and listens to the user in real time, providing feedback to help learners draw parallels between notation and sound.